

·案例报道·

针刀针刺治疗中并发急性咽部出血死亡1例(英文)

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Acute Pharyngeal Hemorrhage as a Fatal Complication Arising from Acupotomy with Needle-knife: A Case Report

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Introduction

Acupotomy medicine, derived from the acupuncture treatments, serves as a relatively new branch of Traditional Chinese Medicine (TCM), which has been proved effective in healing diseases of various origins. Acupotomy can be defined as a process between the operative and non-operative procedure. And the basic theory of its mode of action is an integration of TCM with surgical anatomy. Needle-knife, the core instrument of acupotomy, takes the form of an acupuncture-moxibustion needle with a small scalper on the pinpoint. Since invented in 1970s, needle-knife treatment has been widely practiced in China and the other Southeast Asian countries. Acupotomy is clinically applied to cervical and lumbar spondylosis, osteoarthritis, soft tissue injuries, traumatic bursitis, tenosynovitis, muscular fasciitis and neuralgia. However, this therapy is still not free from complications. In the current report, we described an autopsy case of acute pharyngeal hemorrhage as a fatal complication after acupotomy treatment.

Case Report

Clinical history

A 49-year-old man went to a village clinic for chronic pharyngitis. The therapist, proved illegal for medical practice afterwards, administered him 3.0 g ceftriaxone sodium, 5.0 g dexamethasone and 40 mL

Shuanghuanglian injection (one kind of Chinese patent medicine for pharynx disease) as intravenous drip, which lasted for nearly 15 minutes. Afterwards, the therapist punctured the hyperplastic lymphoid follicles of posterior wall of the pharynx, the hypertrophic lateral pharyngeal bands and other inflamed areas through the mouth with the needle-knife (Fig. 1). During the procedure, the patient developed dyspnea and other symptoms. In spite of this, tracheotomy was immediately carried out, and the patient came to death a few minutes later.



Fig. 1 Standard specifications of the needle-knife

Autopsy findings

The forensic autopsy was performed five days after death. As indicated by autopsy, petechial hemorrhages were seen on the left palpebral conjunctiva and the right lower palpebral conjunctiva. A 2.0 cm surgical incision was observed on the left side of the skin above the thyroid cartilage, surrounded by hemorrhages of muscle and fascia. There was a 3.0 cm surgical incision in the left thyroid gland; a 4.0 cm×1.5 cm hemorrhagic region in the left side; a 3.6 cm longitudinal incision in the trachea lower left of the thyroid cartilage; a 0.3 cm rupture on the left of the pharynx; and a 0.1 cm rupture on the middle of epiglottic cartilage. Congestion and edema could be

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seen in the epiglottis and the root of tongue. Pharynx mucosa and lateral pharyngeal bands were thick and congestive with lympho-follicular proliferation in the posterior wall of pharynx. Major and segmental bronchi were filled with a large gush of bloody fluid (Fig. 2). In both sides of the interlobular serosa were observed lamellar hemorrhages; and in the epicardial areas, petechial hemorrhages. The liver, with increased resistance on cutting, had micro nodules on both the surface and in the sections. A markedly enlarged spleen weighted 280 g.

Microscopic examination

Microscopically, encephaledema was observed to be combined with neuronal degeneration and astrocytes swelling. The cardiac papillary muscle had focal contraction band necrosis. In partial interlobular septa and lumen of bronchioles considerable alveolar edema and congestion, as well as hemorrhages were found (Fig. 3). Interstitial hemorrhages and epithelial necrosis of thyroid gland were transparent, accompanied by large patchy hemorrhage and leukocyte infiltration within the capsule and ambient skeletal muscles. Findings were reported of regularly sized nodules with thin collagen fibers between the hepatocytes while small regenerative nodules with collagen fibers took shape. The examination of coronary arteries and conduction system showed no abnormal changes.

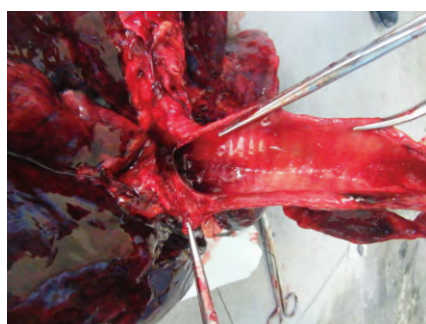


Fig. 2 A large gush of bloody fluid in the major and segmental bronchi

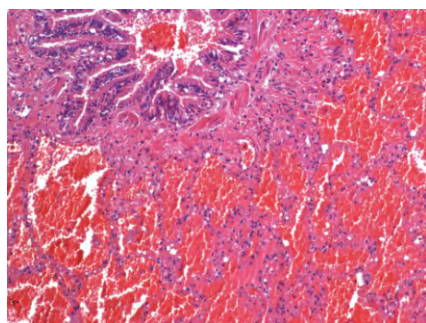


Fig. 3 Hemorrhages in partial interlobular septa and lumen of bronchioles HE×100

Discussion

Acupuncture, originating in China and prevalent

in other Asian counties, is an ancient approach to curing diseases, the complications of which can be traced back to Huang Di Nei Jing (translated as Yellow Emperor's Canon of Internal Medicine) in the Spring and Autumn Period and Hsi Yuan Lu (translated as Washing Away of Wrongs) in the Song Dynasty of China as death caused by the needle performed in a wrong spot. Since acupuncture was introduced into the western countries, multifarious complications have been repeatedly documented, tension pneumothorax reported to be the most probable^[1-3], hemorrhagic complications, acute spinal subdural hematoma^[4], intracranial hemorrhage^[5-6], cardiac tamponade^[7], aortoduodenal or arteriovenous fistulas^[8-9] and pseudoaneurysm of the popliteal artery^[10] and infectious complications such as viral and bacterial infections^[11-12], which have generated diversified consideration for clinic and forensic attention. The accidents caused by acupuncture are continually presented for three reasons: The first reason is that the secure depth of needling is not so easy to master for a beginner, and when the depth surpasses the safe range, organic or vascular lesions are prone to occur. The next reason is that physical presence of the vital organs at some specific acupuncture points can cause such serious problems as respiratory and circulatory failure due to mistreatment of the brain stem under the YaMen acupoint and the aortic root under the QiMen acupoint. Additionally, infection of exoteric microorganisms introduced by needling instruments with improperly disinfected or even without sterilization is another major factor for acupuncture accidents.

Derived from the acupuncture, needle-knife inherits a quite similar morphological character of acupuncture and moxibustion needle, and also integrates with the surgical scalpel, which effectually realizes the enter way, incision, stripping and release of closed surgery for orthopedic diseases such as chronic injury of soft tissues and hyperostosis. Nevertheless, the dual nature of acupotomy is embodied in the merit of morphologic compatibility and demerit of safety. In actual practice, on the other hand, a needle-knife stabs into the body in the form of a needle but acts as a bistoury in that the manual operational techniques of the peeling lysis consist of stab, propulsion, crosswise and slant swing. Hence, safety parameter due to a potential variety of injuries caused by acupotomy often rests with the needle and the knife. In view of the parallel morphological structure and the technical extension, acupotomy shares such equal chances, which should merit attention and precaution.

In addition, acupuncturists practice acupotomy merely by virtue of their anatomical knowledge and without help of ancillary facilities such as X-ray,

computed tomography, ultrasonography or endoscopy for observation after puncture. In other words, acupotomy treatments would be at stake in defect of steady intellectual capital and operating equipment controlling. Other than physical factors, deficiency of theoretical bases and normative clinical guidelines on the part of unqualified practitioners can serve definitely as hindrances to the healthy development of acupotomy. However, a review of the previous literature presented only one case of spinal-cord injury from a broken needle-knife insertion without forensic pathologic examination^[13].

In the current case, asphyxia and increased hemorrhage due to hepatic cirrhosis were deduced from the macroscopic findings as petechial hemorrhages on eye conjunctivae, heart and visceral pleura, and such microscopic findings as contraction band necrosis and brain edema were tenable; however, the latter were capable of increasing the amount of bleeding to a large extent, hence graver consequences. Furthermore, the bloody fluid-filled tracheal and bronchi corresponding with severe pulmonary edema and focal hemorrhages pointed to the cause of death as asphyxia of blood aspiration during the needle-knife treatment. Vascular rupture was not found, but a traumatic injury as thyroid incision definitely served as a major reason of massive and fast bleeding. Meanwhile, the absence of eosinophile granulocyte, mast cell degranulation and IgE rising ascertained the exclusion of bronchial asthma and allergic shock when there were such symptoms as bronchial mucosa shrinkage and dyspnea.

In conclusion, vascular rupture as a serious, therapeutic complication of acupotomy can be theoretically avoided. Although the pathologic findings were not novel in the current case, we theorize that this current case can provide more topic information about the fatalities related to therapeutic procedures of TCM.

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